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Wave Haptics: Building Stiff Controllers from the Natural Motor Dynamics*

Abstract

Haptics, like the fields of robotics and motion control, relies on high stiffness position control of electric motors. Traditionally, DC motors are driven by current amplifiers designed to hide their electrical dynamics. Meanwhile encoder-based position feedback creates virtual springs. Unfortunately this cancellation-replacement approach experiences performance limits due to sensor quantization, discretization, and amplifier bandwidths.

An alternate approach is presented, noting the inherent inductor-resistor dynamics of the motor are beneficial to the haptic task. Two main insights are followed, which may be utilized independently or preferably in combination. First, the electrical inductance L can serve as a stiffness, providing a natural sensorless coupling between the virtual environment and the user. Second, the electrical resistance R can create a natural wave transformation, providing a robust computer interface between the discrete and continuous time domains. The resulting analog circuit implements a simple voltage drive and can achieve higher stiffness than traditional approaches, especially in the frequency range where human users are most sensitive. A prototype 1-DOF system has been implemented and confirms the promise of this novel paradigm.

KEY WORDS—haptics, wave variables, motor dynamics, motor control, current amplifier

1. Introduction

Haptic perception allows humans to gather a considerable amount of information about the mechanical properties of the objects they interact with. Several studies have analyzed the capabilities of human perception, characterizing our sensitivity to different frequencies of contact dynamics (Shimoga 1993; Darian-Smith 1984). In particular, we can detect haptic stimuli up to about 1 kHz, with a sensitivity peak at about 200 Hz (Daniel and McAree 1998).

For the user to correctly perceive contact with simulated virtual objects, the mechanical design and control architecture of haptic interfaces have to stimulate these frequencies. Commonly, devices with impedance causality are designed to minimize their intrinsic friction and inertia (Massie and Salisbury 1994), while the control algorithms have to guarantee a stable interaction with stiff environments over a frequency range matching the users' sensitivity. In this setting, bandwidth and rate hardness (Lawrence et al. 2000) are common metrics to evaluate the effectiveness of haptic displays.

However, the stable implementation of stiff virtual environments over a wide range of frequencies is still a challenging problem. The interconnection between the physical domain and the virtual environment is traditionally achieved by a digital control loop sensing device position and commanding a feedback force to a current controlled actuator (Koren and Ulsoy 1982). Maximum achievable stiffness, and consequently bandwidth and stability of the haptic interaction, is limited by computational delays (Mahvash and Hayward 2004) and information loss caused by time discretization (Colgate and Schenkel 1994; Gillespie and Cutkosky 1996) and position quantization from encoders (Abbott and Okamura 2005; Diolaiti et al. 2006). These phenomena lead to spurious energy

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generation and eventually instability if not dissipated through the intrinsic friction of the device, controller (Hannaford and Ryu 2002), or damping from the user's grasp. Alternatively, the use of analog position measurements and time continuous feedback (Kawai and Yoshikawa 2004) has been shown to improve the dynamic range of haptic rendering (Colgate and Brown 1994).

In this paper we propose an alternative approach to the control of force feedback devices. First we recognize that a haptic interface acts as a transducer that transfers power between the user and device in the mechanical domain and the virtual environment. The fact that the DC motor itself is a transducer that couples mechanical and electrical domains in a stable fashion is often overlooked in the field of haptics. This assumption is justified by the behavior of current amplifiers, which use an internal feedback loop to speed up the electrical dynamics while rejecting back-EMF effects that are treated as disturbances. On the other hand, Mehling et al. (2005) showed the possibility of using electrical dynamics to increase the apparent viscous friction to improve the stability of haptic devices. Full state estimators can also be used in place of tachometers to derive accurate velocity measurements from the back-EMF voltage (Fiene and Niemeyer 2004).

A careful analysis of motor dynamics shows that the electrical inductance L provides a natural stiffness that can couple the virtual environment to the user without a position sensor. The effect is inherently available at high frequencies and can be exploited in the bandwidth of human perception through a simple analog circuit. It does not suffer from the traditional limitations and can provide a seven-fold stiffness improvement on typical hardware.

To provide a fully functional haptic interface, this analog circuit, which replaces the traditional current amplifiers, still needs a computer interface. To not squander the improvements, the circuit can utilize the electrical resistance R to implement a wave transformation (Niemeyer and Slotine 1997, 2004; Stramigioli et al. 2002). Interfacing in wave variables can be robust to sampling delays, providing a more explicit description of power flows and passivity. Encoder-based position feedback in a digital loop is necessary only at low frequencies to counter any motor parametric uncertainty. This relaxes sensor requirements and improves stability characteristics. The overall approach is illustrated in Figure 1.

The paper is organized as follows. In Section 2 motor dynamics are briefly recalled and the mechanical equivalents of electrical dynamics are derived. Their potential to provide encoderless force feedback and position holding is discussed in Section 3. In Section 4 we present the more general framework of wave variables that, through a simple analog circuit, provides an interface between electrical dynamics and virtual environments necessary for haptic rendering. To evaluate the performance of the proposed approach, in Section 5 we focus on the implementation of stiff contact and free motion. Root locus analysis in Section 6 emphasizes differences and

analogies with the traditional approach, while in Section 7 we discuss the design of a digital control loop to provide robustness to parametric uncertainties. Overall system performance is evaluated in Section 8, where we show how the proposed approach is able to stably render high stiffnesses, especially at high frequency, thus matching the characteristics of the human kinesthetic perception. Finally, the experiments on a simple 1-DOF testbed, presented in Section 9 confirm the theoretical expectations.

2. Interpreting the Motor Dynamics

Though generally ignored under the assumptions of an ideal actuator and perfect current amplification, the electrical dynamics of a DC motor are well known to be:

$$e_A(t) = Ri(t) + L \frac{di(t)}{dt} + e_B(t) \quad (1)$$

$$e_B(t) = k_T \dot{x}(t) \quad (2)$$

where e_A is the applied voltage to the armature circuit consisting of the resistance R , the inductance L and the back-EMF voltage e_B . The mechanical dynamics are given as:

$$m\ddot{x}(t) = F(t) - c(\dot{x}(t)) - F_H(t) \quad (3)$$

$$F(t) = k_T i(t) \quad (4)$$

where m is the rotor inertia, c is the (nonlinear) friction and F_H is the user torque *opposing* the motion $\dot{x}$ of the rotor. The torque constant and back-EMF constant are the same physical parameter and are both denoted by k_T . The dynamic system is illustrated in Figure 2(a) and represented in block diagram form in Figure 2(b).

As stated by (2) and (4), the actuator couples the mechanical and electrical domains. Hence, it is possible to take advantage of the coupling equations to interpret the electrical dynamics in the mechanical domain.

Coupling equations (2) and (4) relate torque to current and voltage to velocity. Since the inductance L sets a current proportional to the integral of the its terminal voltage, it corresponds to a spring of stiffness:

$$K_L = \frac{k_T^2}{L} \quad (5)$$

which generates a torque F proportional to the integral of the differential velocity of its endpoints. Similarly, the resistance R maps into a mechanical viscous damper:

$$B_R = \frac{k_T^2}{R} \quad (6)$$

As shown in Figure 3, the equivalent spring and damper share the same torque F , hence they are connected in series between the rotor inertia and the energy conversion element,

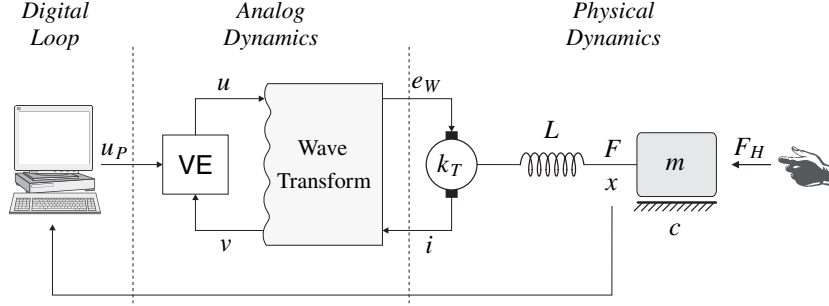


Fig. 1. The wave haptics approach: analog circuitry implements an energy consistent inner loop for stable interaction at high frequency. An external digital loop ensures low frequency accuracy. The motor inductance acts as elastic coupling to the user.

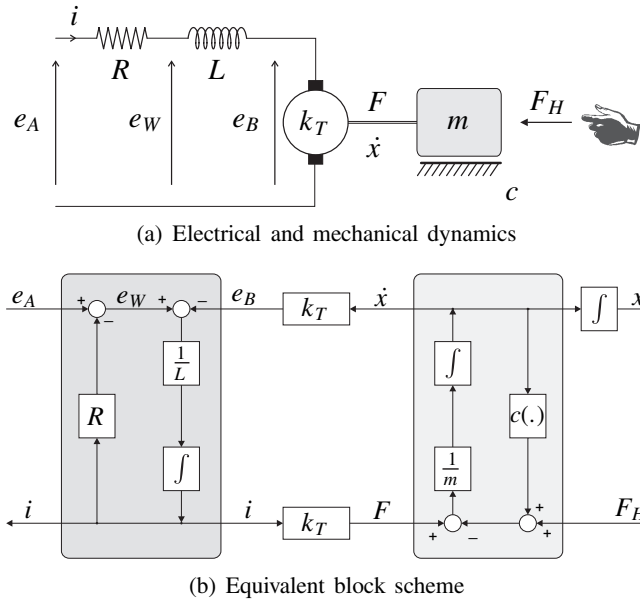


Fig. 2. DC motor dynamics and equivalent block scheme.

which corresponds to a gyrator in bond graph terms (Paynter 1960; Karnopp et al. 2000). Indeed in the language of bond graphs, the inductor and resistor connected to a 1-junction in the electrical domain can be pulled through the gyrator to appear as their mirror elements attached to a 0-junction in the mechanical domain.

Because of the series connection, the motor's behavior is dominated by the damper in the range of frequencies of kinesthetic interaction and the stiffness is often overlooked. A shorted motor feels highly damped, while the stiffness is only apparent for the first few milliseconds.

With low inductance motors commonly used in haptics, K_L can be a very high stiffness. As an example, for the Maxon RE 25 motors found in the PHANToM 1.0 with val-

ues of $k_T = 43.8$ mNm/A and $L = 0.83$ mH, we have $K_L = 2.31$ Nm/rad. With an approximate 8 : 1 gear ratio and lever arm of 14 cm the corresponding tip stiffness reaches 7500 N/m. For comparison, the maximum stable value of a virtual spring implemented according to the traditional scheme has been found to be approximately 1100 N/m (Diolaiti et al. 2006), corresponding to about 0.34 Nm/rad in joint space.

Therefore, a control scheme that treats the motor inductance as a spring in mechanical series and directly controls $\dot{x}_W$ could stably render higher overall stiffnesses than traditional approaches. Indeed K_L is a *physical* element of the system, so it is not affected by the non-idealities of the digital control loop that cause energetic inconsistencies and lead to instability. Moreover, because velocity information is inherently obtained through the back-EMF effect of the motor, this force feedback does not require an actual position sensor and operates naturally at high frequencies.

3. Encoderless Position Holding

We begin with the task of building a simple controller that locks the position of the motor without using explicit position sensing and feedback.

Because we are attempting to directly control $\dot{x}_W$, we consider the inductance L to exist in the mechanical domain as K_L while retaining the resistance R in the electrical domain. This configuration is illustrated graphically and in block diagram form in Figure 4.

This assumption simplifies the electrical dynamics into purely algebraic relationships that can be easily controlled:

$$e_W = e_A - Ri \quad \dot{x}_W = \frac{e_W}{k_T} \quad (7)$$

To this end, it is sufficient to design an analog voltage drive that actively compensates for motor resistance R by setting:

$$e_A = Ri \implies e_W = 0 \implies \dot{x}_W = 0 \quad (8)$$

By using knowledge of the motor's resistance to regulate the voltage across the energy conversion element to zero, we force

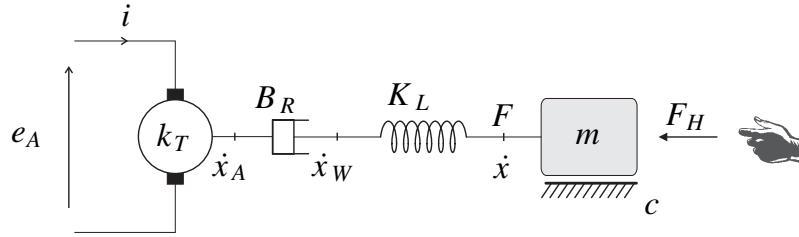
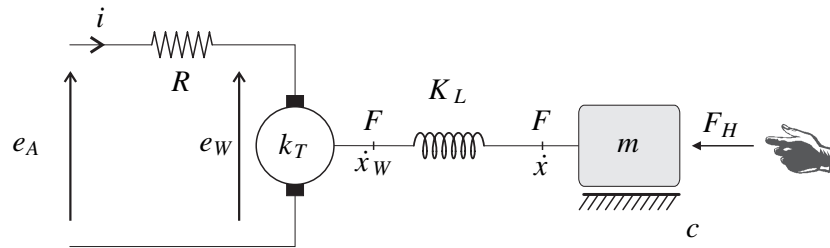
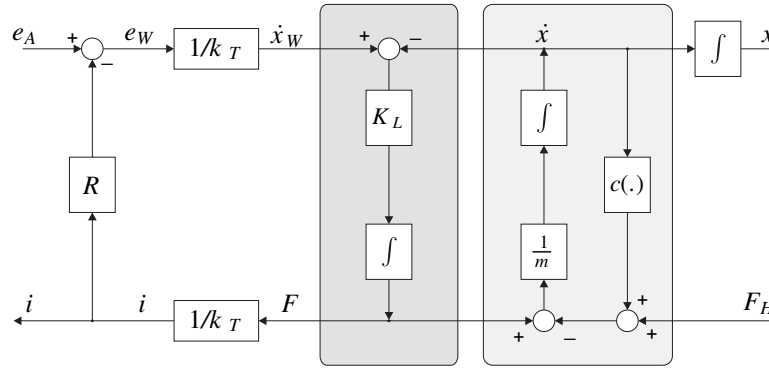


Fig. 3. Rotor inductance L and resistance R interpreted as series connection of a spring $K_L = k_T^2/L$ and a viscous damper $B_R = k_T^2/R$.



(a) System dynamics with the motor inductance in the mechanical domain



(b) Equivalent block scheme

Fig. 4. Interpretation of motor inductance as a mechanical spring K_L for analog circuit design.

the velocity at the spring end-point to zero. We allow the motor inductance to provide the coupling between x_w and the actual rotor position x . The end result is that the user feels a rigid constraint through the high stiffness of K_L .

A possible implementation based on elementary op-amp blocks is depicted in Figure 5. A voltage proportional to the current i is derived by means of a sense resistor R_s and, through a low pass filter that is needed to attenuate high frequency noise, is fed to the compensator that amplifies it by a factor $(R + R_s)/R_s$ (the coefficient α simply scales the actual resistors to values that guarantee a good signal to noise ratio

for the op-amps). The overall transfer function is:

$$E_A(s) = \frac{R + R_s}{R_s} \frac{1}{R_f C_f s + 1} R_s I(s) \quad (9)$$

and at frequencies not attenuated by the filter corresponds to (8), where the total resistance that has to be compensated is $R + R_s$. Finally two complementary power transistors in push-pull configuration are used to apply the commanded voltage to the motor.

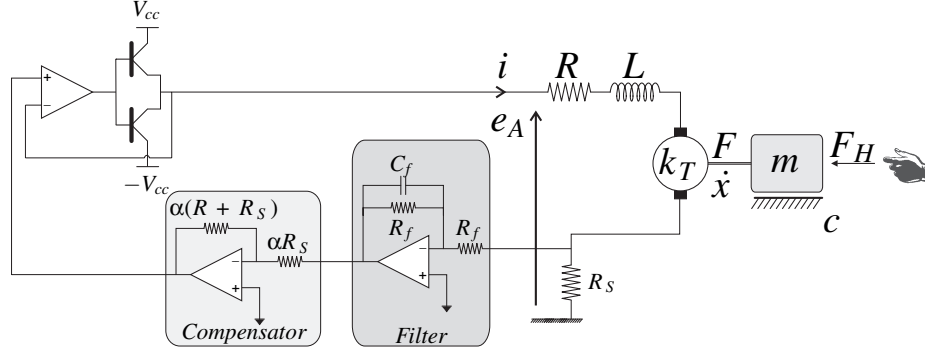


Fig. 5. Diagram of a simple position holding analog circuit.

4. The Wave-haptics Approach

Although providing important insights on the possibility for stiff encoderless force feedback, the simple circuit outlined in the previous section cannot be directly used in haptic interfaces since it lacks the capability of being programmed to render arbitrary virtual environments. On the other hand, electrical currents and voltages provide information about the mechanical domain and such information is fully utilized by the circuit of Figure 5. The current i indicates a measurement of the force F , while the voltage e_w can be reconstructed from e_A through the knowledge of motor resistance to provide a measurement of the velocity $\dot{x}_w$. To implement arbitrary virtual environments, the system has to allow us to program the input–output relation between i and e_w .

Previous studies in the field of telerobotics (Niemeyer and Slotine 1997, 2004), showed that wave variables are a convenient setting to provide robustness to time delays, time discretization, and sampling issues. In view of these considerations, we interpret the motor resistance as part of a natural wave transform, as shown in Figure 6. A wave transform encodes, without loss of information, the normal power variables of velocity and force or of voltage and current into wave variables u and v . According to Niemeyer and Slotine (2004), wave variables are given here by:

$$\begin{aligned} u(t) &:= \frac{e_w + Ri}{\sqrt{2R}} = \frac{B_R \dot{x}_w + F}{\sqrt{2B_R}} \\ v(t) &:= \frac{e_w - Ri}{\sqrt{2R}} = \frac{B_R \dot{x}_w - F}{\sqrt{2B_R}} \end{aligned} \quad (10)$$

The equations on the right express the relationships between wave variables and equivalent mechanical quantities, where $\dot{x}_w(t)$ is the desired motion and F is the common force applied by the spring K_L to the damper and to the rotor inertia m . The equivalent viscous damping B_R serves as the wave impedance.

The change of coordinates introduced by the wave transform (10) allows us to represent the overall instantaneous power $P(t)$ flowing from the virtual environment to the controlled motor as:

$$P(t) = e_w(t)i(t) = \dot{x}_w(t)F(t) = \frac{1}{2}u^2(t) - \frac{1}{2}v^2(t) \quad (11)$$

Each wave variable carries its own power content with $u(t)$ encoding the power flowing into the motor and $v(t)$ representing the power flowing from the motor into the virtual environment. In classical haptic systems, energetic inconsistencies that lead to instability arise because one power variable, typically the feedback force F , is the output of a time-discrete algorithm while the velocity changes during each sampling interval. Here, power flow is individually described by each wave variable independently of the other. System passivity depends only on the magnitudes of the wave variables and is hence unaffected by delays or lags.

To complete the wave transform, the dark shaded part of Figure 6 consisting of the two $\sqrt{2R}$ gains and the summing junction is realized by means of an *analog circuit*. In particular, the output voltage e_A is derived from current sensing as:

$$\begin{aligned} e_A(t) &= \sqrt{2R}u(t) \\ v(t) &= u(t) - \sqrt{2R}i(t) \end{aligned} \quad (12)$$

Because the wave variables $u(t)$ and $v(t)$ exist as electrical signals in the analog circuit, a passive implementation of the virtual environment can be obtained in several ways. First, simple transfer functions $D(s) = U(s)/V(s)$ can be realized in analog hardware. According to Niemeyer and Slotine (1997), their passivity is characterized as:

$$|D(s)| = \left| \frac{U(s)}{V(s)} \right| \leq 1 \quad (13)$$

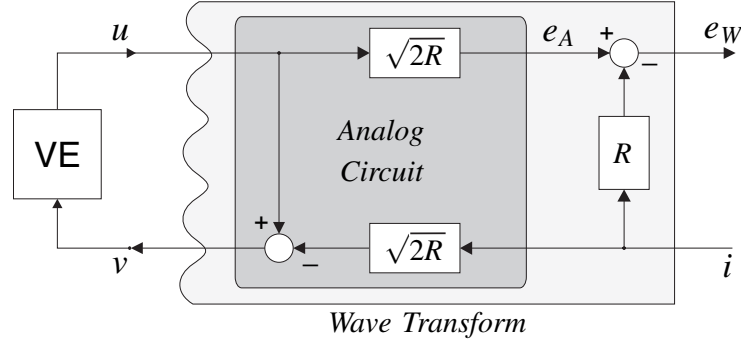


Fig. 6. The wave transform to interconnect the virtual environment (VE) to the electrical domain is implemented by means of an analog circuit.

Alternatively, $v(t)$ and $u(t)$ can be digitized and the virtual environment implemented on a computer either in wave space or in traditional power variables by use of a second decoding digital wave transformation. Energy consistency between the discrete and continuous time domains at the A/D interface is achieved if the sampled value v_h used in the digital virtual environment is such that:

$$T v_h^2 \leq \int_{(h-1)T}^{hT} v^2(\tau) d\tau \quad h \in \mathbb{N} \quad (14)$$

where T is the sampling period. Even though the performance evaluation of different discretization schemes is a matter of future investigation, we note that the condition (14) is satisfied if the integral of the input wave is sampled:

$$v_h = \frac{1}{T} \left| \int_{(h-1)T}^{hT} v(\tau) d\tau \right| \quad (15)$$

which suggests taking advantage of the low-pass characteristics of an anti-aliasing filter to ensure energy consistency.

Passivity is achieved by avoiding energy growth in any wave variable, so that an energetically consistent wave interface guarantees a passive implementation. Moreover, computational delays do not change the energy in a wave variable and hence are also guaranteed not to affect system stability.

5. Haptic Interaction through Wave Variables

By integrating the motor resistance, the wave-haptics circuit of Figure 6 allows us to directly control the velocity $\dot{x}_w$ at the end-point of the spring K_L , as shown in Figure 7. This latter element therefore acts as coupler between the user and the virtual environment implemented in wave variables. Though similar to the virtual coupling concept of Miller et al. (2000),

here the coupling spring has the advantage of being a physical element, so it is not affected by the stability issues of a digital implementation.

The two most demanding virtual environments are free motion and perfectly rigid contact. Both imply an infinite frequency response (i.e., they are purely algebraic systems), as motion occurs immediately for any force in the former and forces are immediately created for any motion in the latter. These two virtual environments constitute the basis of every haptic display. The causality of an impedance device, designed to be light and back-drivable, clearly favors free motion and is challenged by the rendition of rigid contact. Assuming back-drivability, free motion avoids all forces ($F = 0$) and reflects all power carried by the incoming wave $v(t)$ back by means of $u(t)$ as:

$$u(t) = v(t) \iff i = 0 \quad \forall e_w \iff F = 0 \quad \forall \dot{x}_w \quad (16)$$

where (10) is used to convert between wave and power variables. Dually, a rigid contact also reflects all power by suppressing any motion ($\dot{x}_w = 0$):

$$u(t) = -v(t) \iff e_w = 0 \quad \forall i \iff \dot{x}_w = 0 \quad \forall F \quad (17)$$

Note that this does not hold the applied voltage e_A at zero, but only cancels the voltage across the inductor and back-EMF. It does not short the motor; instead it effectively sets $e_A = Ri$ as seen in Section 3. Therefore, in this situation K_L is the maximum stiffness that can be rendered to the user.

5.1. Analog Implementation of the Unilateral Wall

In order to evaluate the performance of the wave-haptics scheme, we use analog hardware to implement the transfer functions corresponding to free space (16) and to rigid contact

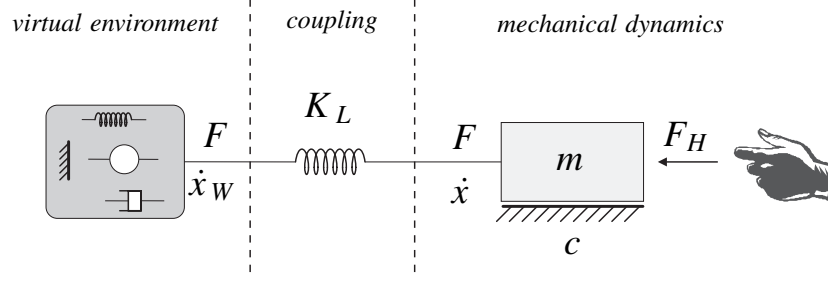


Fig. 7. Motor inductance acts as *physical* coupling with the virtual environment.

(17). A collision detection algorithm based on direct position measurements generates a logical signal

$$\delta_{\text{FREE}}(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases} \quad (18)$$

that commands switching between the two environments to simulate a virtual rigid wall with unilateral constraint:

$$u(t) = \eta(x)v(t) \quad \text{where} \quad \eta(x) = -1 + 2\delta_{\text{FREE}}(x) \quad (19)$$

Note however that the encoder is not used to directly compute the force fed back to the user.

When incorporating and utilizing the electrical dynamics, K_L is the maximum stiffness that can be rendered by means of a passive wave-haptic approach. Disturbances may stem from the unmodeled high frequency dynamics of the power transistors used to achieve the desired voltage input e_A , as well as from other sources of electrical noise in the control loop. However, these phenomena occur at much higher frequencies (several hundred kilohertz) than the perceptual bandwidth of the human operator (about one kilohertz). Consistent with Figure 5, a high frequency low-pass wave filter:

$$H(s) = \frac{\lambda}{s + \lambda} \quad (20)$$

can then be included in series with the virtual environment to address these limitations without significantly affecting the perceived transients (Tanner and Niemeyer 2004). Such a filter attenuates the high frequency content of the wave variables and, because of (11) and (13), it corresponds to frequency dependent power dissipation.

The transfer function $D(s)$ of the overall controller dynamics in the wave domain becomes:

$$D(s) = \frac{U(s)}{V(s)} = \eta(x) \frac{\lambda}{s + \lambda} \quad (21)$$

where the magnitude of η can be further tuned in the interval $-1 \leq \eta \leq 1$ to remove power and introduce damping.

6. Comparison to the Classical Approach

A root locus analysis allows us to compare the stability characteristics of the wave-haptics approach to the traditional scheme built upon current amplifiers and encoder feedback. For both cases, we first examine the electrical dynamics without mechanical motion, then include the mechanical elements. We ignore the dynamics of power transistors and other high-speed electronic components.

A traditional current amplifier aims at speeding up the motor electrical pole while rejecting the back-EMF voltage. Usually a PI controller regulates the current, setting:

$$E_A(s) = k_A \frac{\tau_A s + 1}{\tau_A s} [I_c(s) - I(s)] \quad (22)$$

where $I_c(s)$ is the commanded current set-point (Leonhard 2001). As shown by the root locus of Figure 8(a), the time constant τ_A is chosen to be moderately faster than the motor electrical pole $-R/L$. A positive proportional gain k_A speeds up the overall electrical dynamics, while the integrator ensures good low-frequency tracking. The high gain also rejects the back-EMF effect, which consequently only marginally moves the electrical poles and is not included in the root locus diagram.

On top of the current regulator (22), force feedback is obtained by closing a loop around the measured device position x :

$$F_c(s) = k_T I_c(s) = K_c [X_c(s) - X(s)] \quad (23)$$

where K_c is the assigned stiffness and $X_c(s)$ is the position of the virtual wall. The corresponding root locus, Figure 8(a), inherits the closed-loop electrical dynamics and includes two poles for the second-order mechanical dynamics of (3) with $F_H = 0$. For high force feedback gains K_c the mechanical poles tend to move to the right half-plane. Mechanical friction c (here considered linear for simplicity) becomes crucial to ensure stable operation, even neglecting the fact that (23) is usually implemented as a digital control loop where time

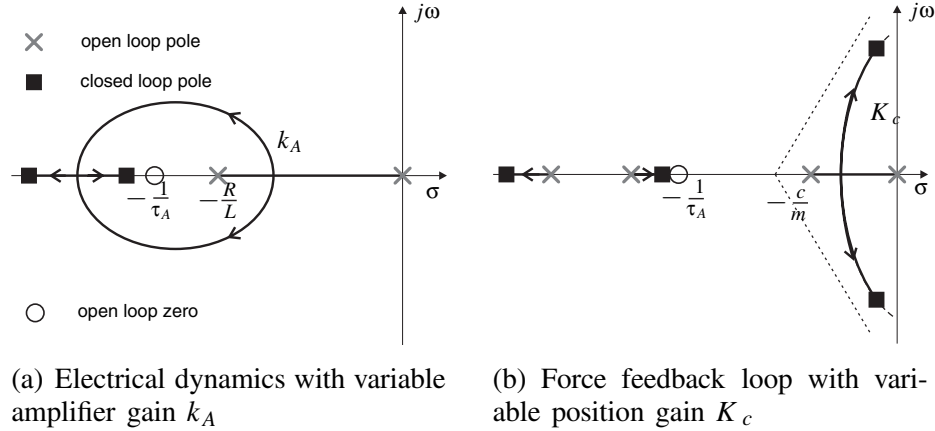


Fig. 8. Root locus interpretation of the classical approach.

discretization and position quantization further limit the maximum achievable gain K_c .

To derive the root locus corresponding to the wave-haptics scheme defined by (12) and (21), we consider the analog circuit of Figure 6 to have a tunable gain R_0 that, in the ideal case, matches motor resistance R . Hence the electrical feedback in the wave-haptics controller is:

$$E_A(s) = -2R_0 \frac{D(s)}{1 - D(s)} I(s) \quad (24)$$

In the case of stiff contact ($\eta = -1$) and with the wave filter (20) this simplifies to:

$$E_A(s) = R_0 \frac{2\lambda}{s + 2\lambda} I(s) \quad (25)$$

As shown in Figure 9(a), the pole related to the wave filter $H(s)$ appears at -2λ . Because of positive feedback used by the wave-haptics circuit, closed-loop poles move outward, with the electrical pole of the motor reaching a pure integrator $1/(Ls)$ as $R_0 \rightarrow R$.

The relatively low gain electrical feedback does not overcome the back-EMF, indeed this effect ensures the position feedback. Introducing the back-EMF loop on top of the current dynamics of Figure 9(a), we include the mechanical pole $-c/m$ relating force to velocity and a zero at -2λ corresponding to the wave filter and relating the back-EMF disturbance to current. The root locus of Figure 9(b) gradually raises the back-EMF constant k_T^2 from zero to its nominal value and places the dominant second-order poles according to the inductor's stiffness k_T^2/L . The wave filter $H(s)$ moves the oscillatory poles and asymptotes slightly to the left, improving their stability regardless of mechanical friction. The root locus also agrees with the passivity of the approach, as all passive systems have at most relative degree 2 with their asymptotes

in the left half-plane and are stable in feedback regardless of gain. For comparison, the traditional loop closure has relative degree 3 and is therefore more difficult to stabilize for high gains K_c . Finally note that an inexact estimate of the electrical resistance R will adjust the closed-loop poles, with a detailed description following in Section 7.

In the case of free space ($\eta = 1$), the electrical feedback (24) simplifies to:

$$E_A(s) = -2 \frac{R_0 \lambda}{s} I(s) \quad (26)$$

and regulates the current to zero. Compared to the PI controller (22), it may leave the electrical dynamics more oscillatory but with a settling time of no more than twice the uncontrolled value. And though this behavior appears at a frequency beyond the human perceptual bandwidth, it can be minimized with a suitable choice of the time constant λ of the wave filter.

7. Robustness and Absolute Position Feedback

Because of thermal effects during motor operation, R can increase by about 20% of its nominal value, thus making an exact calibration and subsequent tuning difficult in practice. This affects both the simple circuit of Figure 5 and the wave transformation circuit of Figure 6. Both fundamentally implement the same idea and require knowledge of R .

As suggested by the previous root locus analysis, robustness and passivity of the control scheme are retained only if the estimated value R_0 closely matches the actual resistance R , leaving a small residual ρ . In mechanical terms, the residual electrical resistance ρ maps into a viscous damper:

$$B_\rho = \frac{k_T^2}{\rho} \quad (27)$$

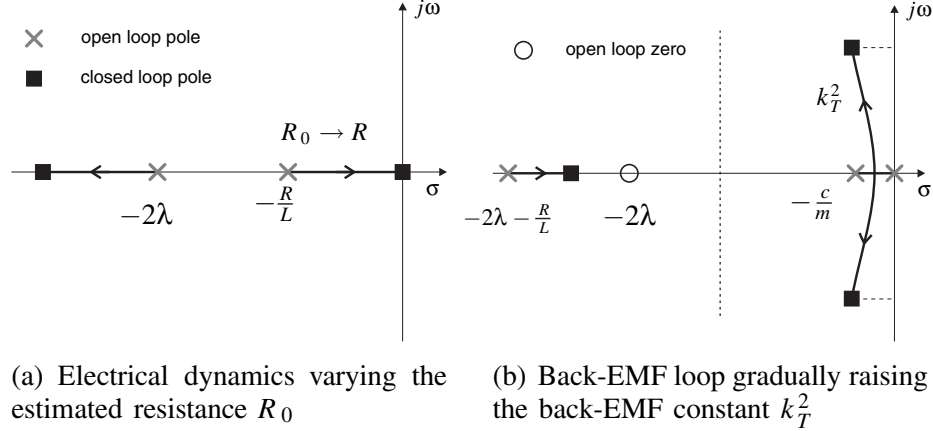


Fig. 9. Root locus interpretation of the wave-haptics controller for stiff contact ($\eta = -1$).

that is connected in series with the spring K_L . This situation is sketched in Figure 10, where we note that the velocity $\dot{x}_\rho$ of the spring endpoint differs from the velocity $\dot{x}_w$ controlled by the wave-haptics circuit:

$$\dot{x}_\rho = \dot{x}_w - \frac{1}{B_\rho} F \quad (28)$$

Although the system is able to exploit the stiff behavior at frequencies lower than the natural frequency R/L of the motor electrical pole, the presence of the residual damper B_ρ is responsible for viscous behavior at frequencies up to $\rho/L < R/L$. At these frequencies, the spring endpoint x_ρ may slowly drift from the desired position x_w . Note, over-estimation and over-compensation with associated negative values of ρ leads to an active residual element. In this case, the endpoint may slowly drift against the user instead of with the user. In both cases the following solution prevents such drift.

Recovering the cascaded loop closure outlined in Figure 1, absolute position feedback is used in a digital control loop to provide low frequency performance and drift cancellation during contact with a stiff virtual object. Since motor inductance L and torque constant k_T are less temperature sensitive than the resistance R , an estimate of the spring end-point position x_ρ can be derived as:

$$x_\rho = \frac{L}{k_T} i + x \quad (29)$$

where i and x are gathered through position and current sensors. A wave signal $u_p(t)$ is computed by a discrete time algorithm according to the simple proportional law:

$$u_p = P [x_{sp} - x_\rho] \quad (30)$$

where P is the proportional gain and x_{sp} is the desired position for the endpoint x_ρ . Finally u_p is summed to the output of the

analog circuit (12) to obtain:

$$U(s) = D(s)V(s) + H(s)U_p(s) \quad (31)$$

By comparing (30) to (10), the compensation wave can be interpreted as an elastic force that is added to F in (10) in order to guarantee that x_ρ reaches, at least in steady state, the desired position x_{sp} . And with reference to (12), the wave signal $u_p(t)$ alters the applied voltage e_A , thereby moving x_A and x_w . A similar effect could be achieved in the simple circuit of Figure 5 adjusting e_A , though without the passivity and robustness understanding afforded by wave variables.

As noted in the introduction, humans are particularly sensitive at frequencies about 200 Hz, while at lower frequencies the poor position resolution of the neuromuscular apparatus limits our ability to identify the rendered stiffness. Therefore the feedback gain P , which only alters the low-frequency rendering, can be kept relatively small. This avoids many of the stability issues of digital implementation (Diolaiti et al. 2006) and allows us to ignore the associated non-idealities in the following performance evaluation.

8. Performance Evaluation

To evaluate the system, we assume $x_{sp} = 0$ and neglect the wave filter $H(s)$, which is used to counter electronic noise without significantly affecting performance in the frequency range of haptic interaction. Previous considerations allow us to approximate the outer digital loop as a time continuous system.

In the case of stiff contact ($\eta = -1$), the transfer function between the user input force $F_H(s)$ and the feedback force $F(s)$ is described as:

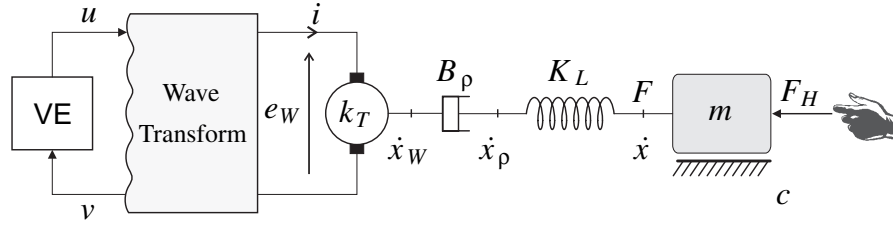


Fig. 10. The uncompensated residual resistance ρ maps into a mechanical damper B_ρ . The wave circuit does not directly control the spring endpoint x_ρ .

$$\frac{F(s)}{F_H(s)} = \frac{K_L B_\rho \left(s + P \sqrt{\frac{R_0}{2k_T^2}} \right)}{B_\rho \left(s + P \sqrt{\frac{R_0}{2k_T^2}} \right) (ms^2 + cs + K_L) + s K_L (ms + c)} \quad (32)$$

while device position $X(s)$ is described as:

$$\frac{X(s)}{F_H(s)} = - \frac{B_\rho \left(s + P \sqrt{\frac{R_0}{2k_T^2}} \right) + K_L}{B_\rho \left(s + P \sqrt{\frac{R_0}{2k_T^2}} \right) (ms^2 + cs + K_L) + s K_L (ms + c)} \quad (33)$$

where c is the linearized mechanical friction. Therefore, the frequency dependent stiffness $K(s)$ is:

$$K(s) = - \frac{F(s)}{X(s)} = \left[\frac{1}{K_L} + \frac{1}{B_\rho \left(s + P \sqrt{\frac{R_0}{2k_T^2}} \right)} \right]^{-1} \quad (34)$$

In other words, the total stiffness $K(s)$ is given by the parallel connection of the electrical stiffness K_L and the spring generated by the digital controller. In particular, in the case of ideal compensation ($\rho \rightarrow 0$ and $B_\rho \rightarrow \infty$), the electrical stiffness K_L is active at all frequencies. On the other hand, it is clear that without the digital controller ($P = 0$), the residual viscous damping B_ρ dominates at low frequencies, causing the position (33) to drift. This phenomenon is finally compensated by the gain P , that, together with the wave gain R_0 , determines the bandwidth and the stiffness of the low-frequency spring:

$$K_p = B_\rho \sqrt{\frac{R_0}{2k_T^2}} P = \frac{B_\rho}{\sqrt{2B_0}} P \quad (35)$$

Note that this stiffness depends on B_ρ and is sensitive to parametric uncertainties on ρ . Previous considerations of user perception suggest, however, that such uncertainty can be tolerated.

The behavior of the wave-haptics controller can be best viewed by means of Bode plots. In particular, Figure 11 compares the stiffness (34) rendered by different control schemes

in the case of rigid contact. Plots are derived from the parameters of the Maxon RE25-118743 motor used for experiments in the next section. Although neglected in the analytic derivations, a wave filter (20) with $\lambda = 100,000$ rad/s is considered in the plot.

First, the natural behavior $K_{\text{natural}}(s)$ of a DC motor when $e_A = 0$ (solid thin line) exhibits, as expected, a flat behavior with magnitude K_L at high frequency, normally beyond the bandwidth of human kinesthetic perception. Lower frequencies are dominated by electrical dissipation.

When stiff contact (17) is simulated, the analog circuit (12) actively compensates for motor resistance. As shown by the curve K_{wave} (dashed) with $R_0 = 0.9R$, the analog circuit shifts the region of high stiffness to the range of frequencies corresponding to peak human sensitivity. Residual low frequency attenuation is then compensated by the proportional controller (30), as illustrated by the curve $K_{\text{wave}+P}$.

For comparison, the dash-dotted line represents the behavior of the traditional haptic controller as described by (22) and (23). The curve labeled $K_{\text{classical}}(s)$ shows how current amplifiers are designed to speed up the motor electrical pole while rejecting back-EMF through a high feedback gain eventually provided by an integrator. By recalling (3), the digital loop directly commands the feedback force on the basis of the measured position displacement. Hence in the traditional scheme the region exhibiting electrical stiff behavior is shifted to an even higher frequency. The digital force feedback loop, approximated here as a continuous-time system, is thus required to provide stiff behavior over the whole frequency range of human kinesthetic perception. When high stiffnesses are rendered, quantization and discretization effects become non-negligible at frequencies about the human sensitivity peak and stability issues arise (Diolaiti et al. 2006).

The advantage of the wave-haptics approach is therefore to use digital control only in a range of frequencies that falls within its capabilities with the usual servo rates and position resolution. The physical spring K_L naturally provided by the motor is retained to achieve stable stiff kinesthetic interaction at high frequencies.

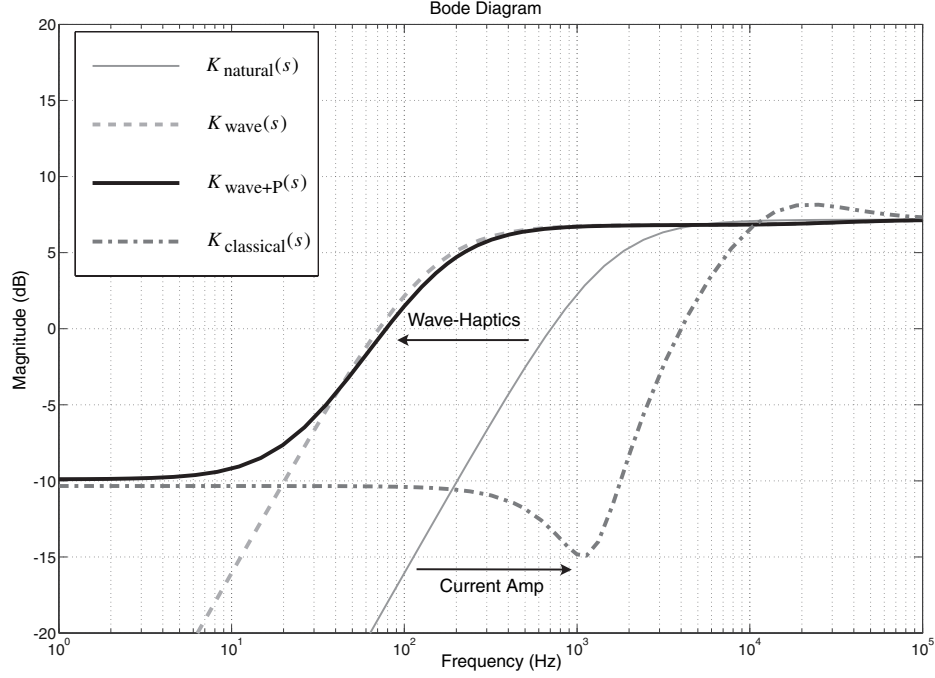


Fig. 11. Bode diagrams comparing the wave-haptics and the classical controllers in the case of stiff contact.

The corresponding Bode diagrams for free motion ($\eta = 1$) are illustrated in Figure 12. Analogously to Figure 11, a wave filter (20) with $\lambda = 100,000$ rad/s is considered to derive the plots. In this case the difference between the wave-haptics scheme and the traditional controller is relatively modest; both approaches build a current loop that contains an integral effect. On the other hand, the presence of a proportional term in the traditional controller (22) filters the resonant peak that otherwise appears in the wave-haptics plots. This effect is located at frequencies well beyond human perceptual capabilities and experiments suggest little impact. Further investigations are required to optimize wave filter design for minimizing the resonant peak. Finally, note that the curves K_{wave} and $K_{\text{wave}+P}$ are almost coincident, thus the proportional controller (30) can be turned off when simulating free space.

9. Implementation and Experiments

We have realized a simple prototype system that extends the scheme of Figure 5, allowing us to implement the control equations (21), (30) (31) with $\lambda = 100,000$ rad/s, $\eta = \pm 1$. The block diagram of the overall system is sketched in Figure 13.

In particular, the user grasps a handle mounted at the shaft of a Maxon RE25-118743 motor that features $K_L = 2.2815$ Nm/rad, similar to the electrical stiffness of motors commonly found in PHANTOM devices. Power amplification is performed by a pair of complementary MOSFET transistors

in a push-pull configuration and an analog switch alternates between free space ($\eta = 1$) and rigid contact ($\eta = -1$). Data is collected at 5 kHz using an A/D converter to measure motor current through a sense resistor and a high resolution encoder (10^4 counts per revolution) to measure the motor shaft position. The proportional compensation (30) and the collision detection (19) are computed using a more realistic servo rate of 1 kHz and a position resolution of 2,000 counts per revolution. The location of the virtual wall has been set to $x_{sp} = 0$ for simplicity and contact is experienced for $x \geq 0$.

Preliminary results obtained for repeated impacts against the virtual wall are shown in Figure 14 and 15. In these experiments the wave gain R_0 has been tuned to match the motor resistance R with an error of less than 2%, while the proportional controller has been disabled (i.e. $P = 0$). To limit thermal effects, current has been limited at the power supply to 1A. From Figure 14 we see the circuit correctly renders the behavior of free space, since when $x < 0$ the feedback torque F is zero. The voltage e_A adjusts slightly to counter the back-EMF voltage. As contact is experienced, current and torque quickly rise to their maximum value before saturation intervenes at about $F \simeq 0.021$ Nm. Better viewed in Figure 15, the stiffness rendered during the compression phase is approximately $K \simeq 1.9$ Nm/rad, in good agreement with the expectations from the previous analysis.

We note that the compression and the restitution phases appear asymmetric. This behavior is a direct consequence of voltage saturation, which is not yet included in the simple

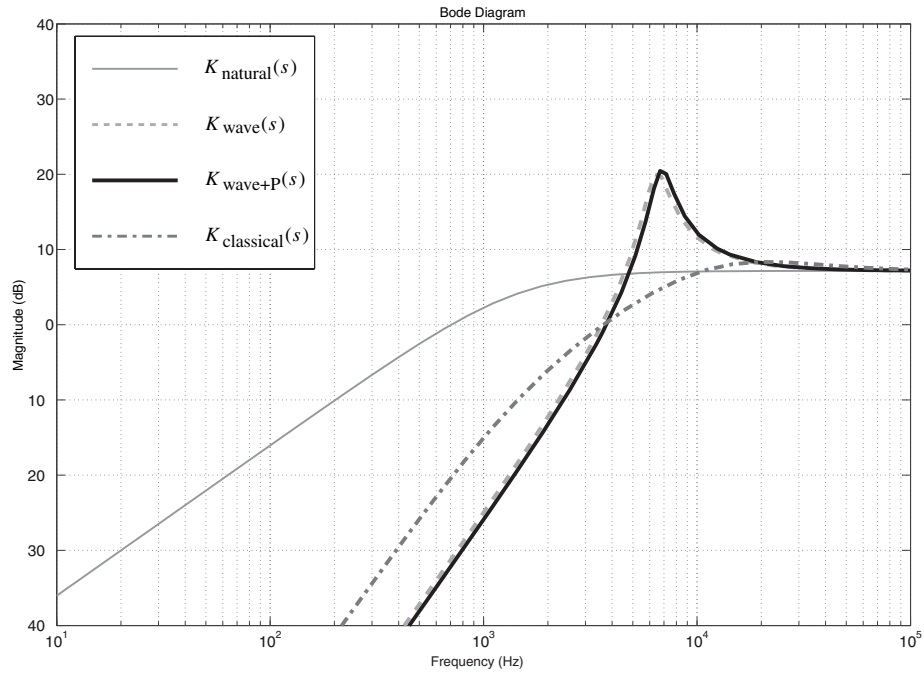


Fig. 12. Bode diagrams comparing the wave-haptics and the classical controllers in the case of free motion.

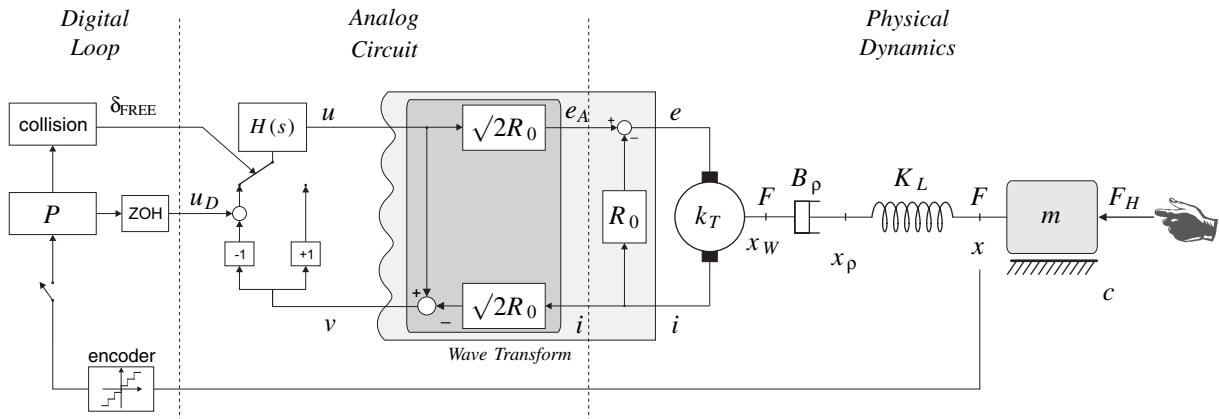


Fig. 13. The complete scheme for virtual wall simulation. Collision detection drives an analog switch that selects between free space and stiff contact. In this latter case, proportional digital controller compensates for residual damper B_ρ . The resulting voltage e_A is applied to the motor.

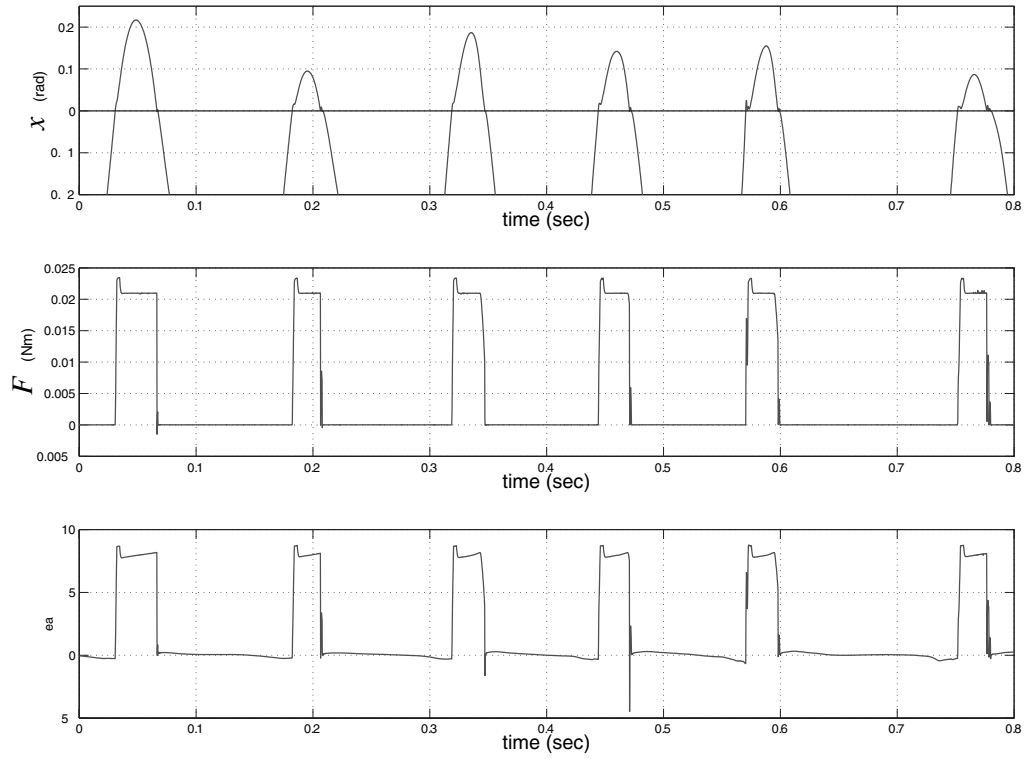


Fig. 14. Repeated contacts with the virtual wall: position, force and voltage diagrams— $P = 0$.

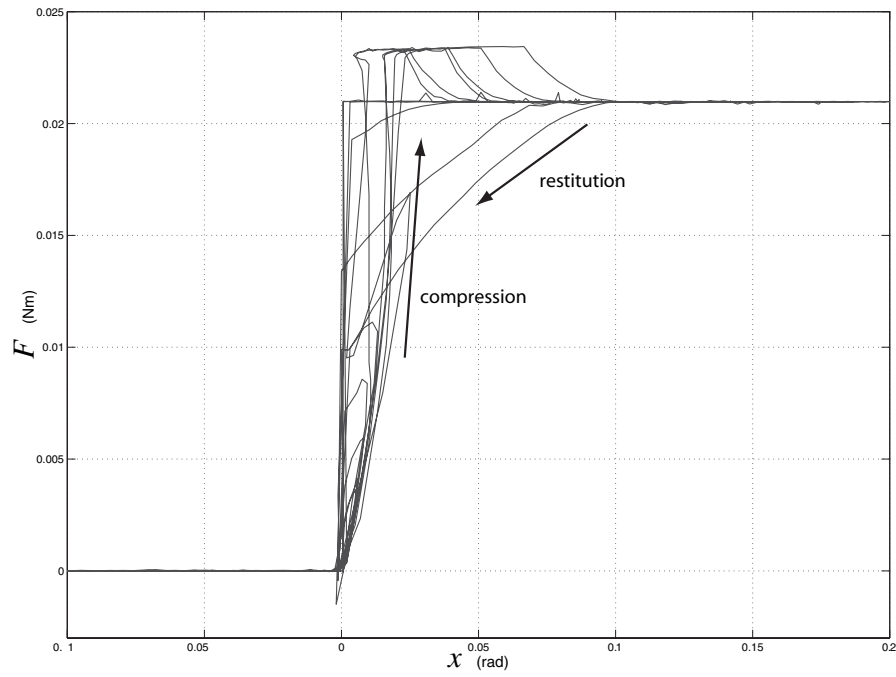


Fig. 15. Behavior of the feedback torque F during contact with the virtual wall - $P = 0$.

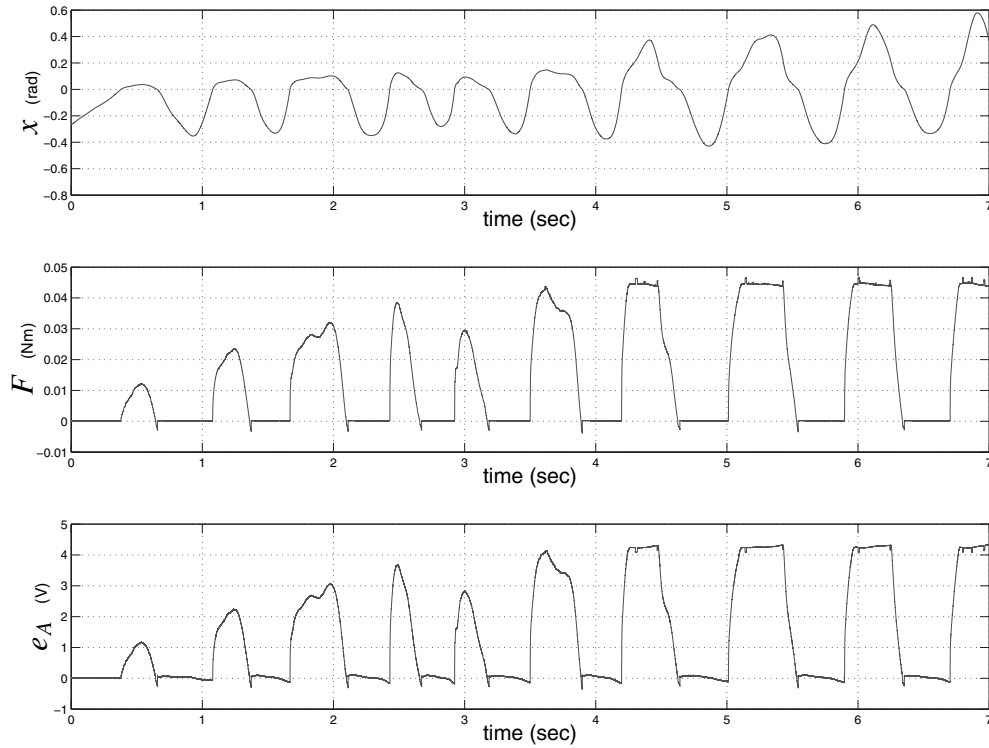


Fig. 16. Repeated contacts with the virtual wall: position, force and voltage diagrams— $K_p \simeq 0.02$ Nm/rad.

dynamics (1). When the drive voltage hits a fixed saturation limit, the back-EMF effects cannot be properly canceled and the current i is affected by motion. As the compression slows, the resisting force drops to its steady state value and, as motion begins during restitution, the restoring force drops accordingly. Effectively the spring forces are overlaid with the back-EMF's viscous damping.

Similarly to what was noted in Section 7, the voltage modification caused by the saturation also shifts the endpoint x_ρ of the spring K_L , since the voltage e_A is not sufficient to achieve perfect compensation of the term Ri . Fortunately, as contact is broken, the behavioral switch via η resets the system for the next collision.

Additional experiments have been repeated in a more realistic situation, by tuning $R_0 \simeq 0.9R$ and taking advantage of absolute position feedback. Moreover, current saturation has been increased to 1.85 A. In particular, results obtained with a low proportional gain of $K_p \simeq 0.02$ Nm/rad are presented in Figure 16 and Figure 17. By looking at the latter diagram, we notice that the residual motor resistance causes a hysteresis loop to arise. The area encircled by compression and restitution phases corresponds to energy dissipated by the residual damper. Moreover, the compression phase is characterized by

two different behaviors. Immediately after impact occurs, the high frequency stiffness provided by K_L dominates. Because of the proportional controller, the rendered stiffness decreases as a quasi-static regime is reached, with the user slowly moving the handle further into the virtual object.

The corresponding diagram obtained with $K_p \simeq 0.38$ Nm/rad is shown in Figure 18. By increasing the gain of the digital controller we notice that compression and restitution phases generate almost overlapping diagrams. Little energy is lost and the residual damper is properly counteracted by the proportional controller. Moreover, during the compression phase the digital controller is sufficient to prevent the spring endpoint x_ρ from drifting from the desired position. The rendered stiffness is in this case of about 2 Nm/rad, in good agreement with theoretical expectations and about 7 times higher than that obtained through traditional current amplification. After saturation occurs, the proportional controller enables the restitution phase to take place at the correct location.

A small negative force can be seen in all Figures 15, 17, and 18. This occurs because both the position controller and the collision detection response is slightly delayed by the digital implementation. The user temporarily sticks to the surface

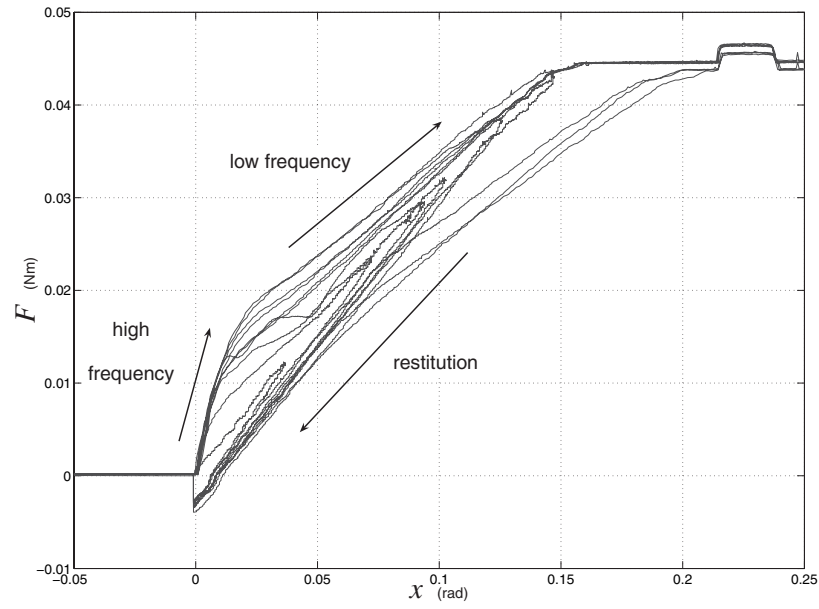


Fig. 17. Behavior of the feedback torque F during contact with the virtual wall— $K_p \simeq 0.02$ Nm/rad.

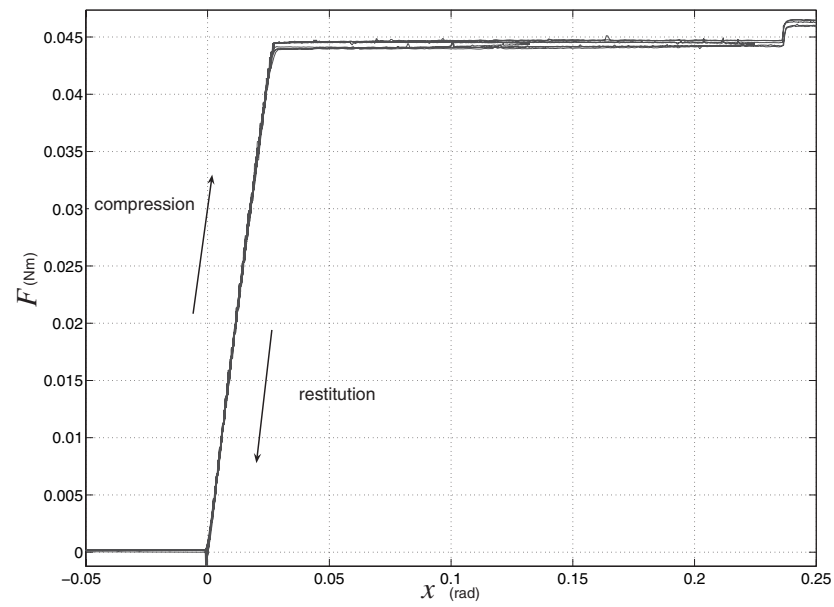


Fig. 18. Behavior of the feedback torque F during contact with the virtual wall— $K_p \simeq 0.38$ Nm/rad.

as the system switches from contact to free space. The effect is being addressed in current work, rectifying the wave signal and implicitly ensuring that applied forces are unidirectional out of the surface.

As a final remark, both Figure 17 and Figure 18 exhibit a small discontinuity at about 0.22 rad. It corresponds to a motor brush commutation that is not filtered through the analog circuitry. Such discontinuities may be perceived by the user and the application to brushless motors promises smoother operation.

10. Conclusions

In this paper we propose to exploit the electrical dynamics of DC motors to improve force feedback for virtual reality and telerobotic applications. With the motors commonly used in haptic displays, the equivalent stiffness provided by the low motor inductance is higher than the stiffness that can be achieved by means of a classical digital control loop and provides force feedback without requiring direct position sensing.

An analog circuit has been designed to replace traditional current amplifiers in order to exploit this effect. Rather than speeding up the electrical dynamics, the circuit only slows down the natural response. It uses a voltage drive, effectively moving the set-point of the equivalent inductance spring. Wave variables are available as analog signals, facilitating the implementation of a passive and power-consistent computer interface and coupling to virtual environments.

As the physical inductance-based stiffness is available at all frequencies, the system is able to improve rendering over a frequency range matching critical human perceptual characteristics. The need for a digital control loop is shifted to a lower frequency range, where quantization and discretization do not substantially affect system stability.

This approach is very appealing with its intrinsic simplicity and its better use of the physical components of the haptic device. It does not require assumptions about mechanical friction to ensure stability. Conversely, the passivity is obtained constructively and the effects of non-idealities are confined behind the wave variable transformation, guaranteeing intrinsic robustness to servo delay.

The realization of more complex, multi-dimensional virtual environments will require an examination of different discretization techniques for the analog wave signals as well as the coordination of multiple motors. We hope this work has laid a strong foundation for such developments and will simplify haptic devices and improve their performance.

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